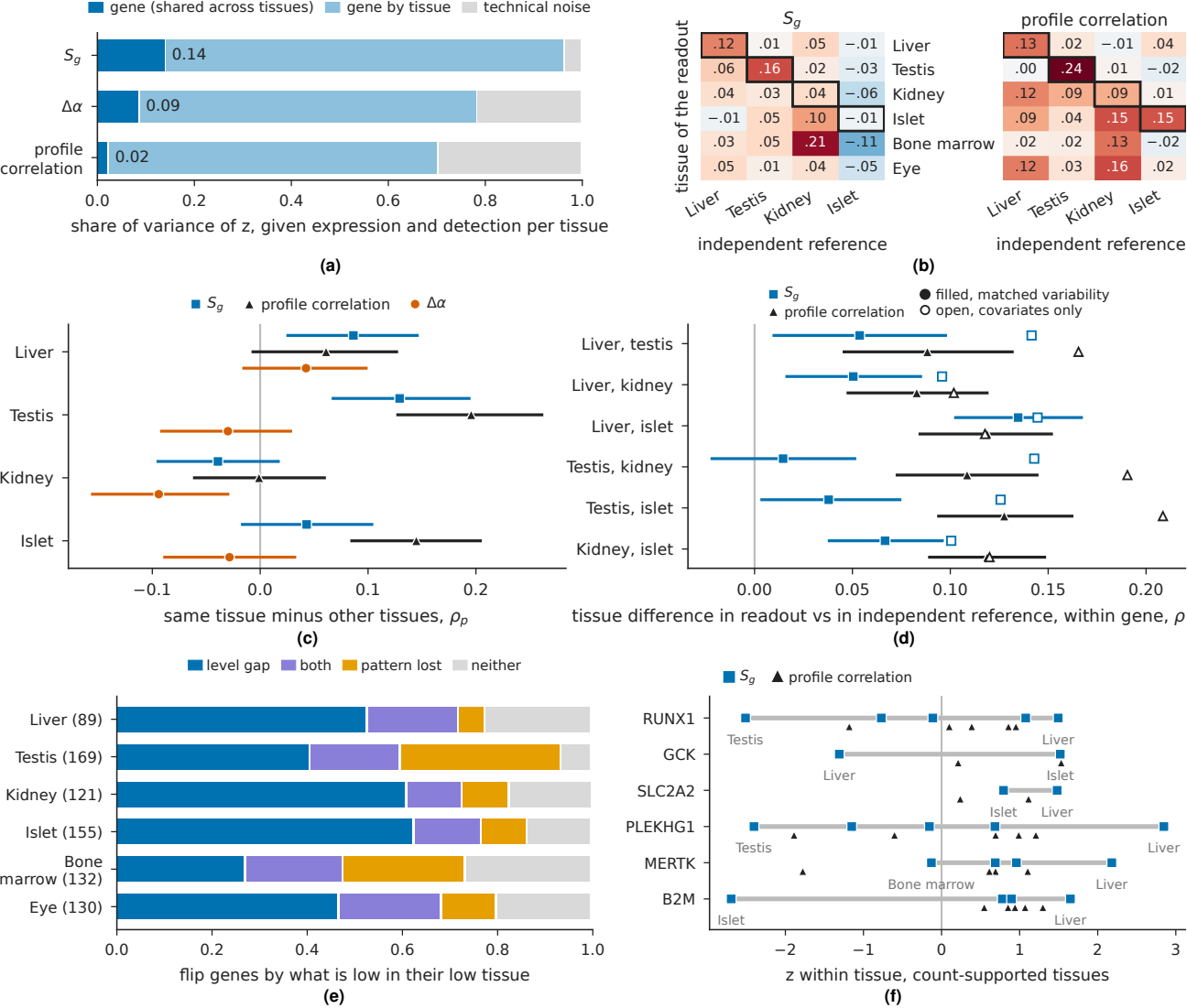

gene (shared across tissues)
gene by tissue
technical noise
S_g
$\Delta\alpha$
profile correlation
0.14
0.09
0.02
0.0
0.2
0.4
0.6
0.8
1.0
share of variance of z, given expression and detection per tissue
(a)
S_g
tissue of the readout
.12 .01 .05 −.01 Liver
.06 .16 .02 −.03 Testis
.04 .03 .04 −.06 Kidney
−.01 .05 .10 −.01 Islet
.03 .05 .21 −.11 Bone marrow
.05 .01 .04 −.05 Eye
Liver
Testis
Kidney
Islet
independent reference
profile correlation
.13 .02 −.01 .04
.00 .24 .01 −.02
.12 .09 .09 .01
.09 .04 .15 .15
.02 .02 .13 −.02
.12 .03 .16 .02
Liver
Testis
Kidney
Islet
independent reference
(b)
S_g
profile correlation
$\Delta\alpha$
Liver
Testis
Kidney
Islet
−0.1
0.0
0.1
0.2
same tissue minus other tissues, ρ_p
(c)
S_g
profile correlation
filled, matched variability
open, covariates only
Liver, testis
Liver, kidney
Liver, islet
Testis, kidney
Testis, islet
Kidney, islet
0.00
0.05
0.10
0.15
0.20
tissue difference in readout vs in independent reference, within gene, ρ
(d)
level gap
both
pattern lost
neither
Liver (89)
Testis (169)
Kidney (121)
Islet (155)
Bone marrow (132)
Eye (130)
0.0
0.2
0.4
0.6
0.8
1.0
flip genes by what is low in their low tissue
(e)
S_g
profile correlation
RUNX1
Testis
Liver
GCK
Liver
Islet
SLC2A2
Islet
Liver
PLEKHG1
Testis
Liver
MERTK
Bone marrow
Liver
B2M
Islet
Liver
−2
−1
0
1
2
3
z within tissue, count-supported tissues
(f)